

**BRAC UNIVERSITY**  
**Department of Computer Science and Engineering**  
**CSE 260: Digital Logic Design**

Examination: **Quiz 1**  
 Duration: 25 Minutes

Semester: Summer 2025  
 Full Marks: 15

Answer the following questions. You **MUST** show your workings/calculations where applicable.  
 Figures in the right margin indicate marks.

|                       |          |     |
|-----------------------|----------|-----|
| Name: <i>Solution</i> | Section: | ID: |
|-----------------------|----------|-----|

1. Perform the following conversion:  $(101011110001100)_{\text{Excess-5}} = (?)_{12}$   
 [CO1] Note: You **must** show all the necessary conversions. [3]
2. Subtract  $(-36)_{10}$  from  $50_{10}$  using 7-bit 2's complement system (show the conversion first).  
 [CO1] Justify whether there is an **overflow or not**. [6]
3. Multiply  $(32112)_5$  with  $(143)_6$ . Perform the multiplication in base-5 and make sure your final answer is in base-5.  
 [CO1] Note: You **must** show the necessary calculations. [6]

$$\begin{array}{cccccc}
 \underline{1} & & \underline{0101} & \underline{0111} & \underline{1000} & \underline{1100} \\
 & & 5 & 7 & 8 & 12 \\
 & & \downarrow & \downarrow & \downarrow & \downarrow \\
 & & (0 & 2 & 3 & 7)_{10}
 \end{array}$$

$$(237)_{10} = (?)_{12}$$

$$\begin{array}{r}
 12 \overline{) 237} \\
 \underline{12} \phantom{0} \phantom{0} \\
 12 \overline{) 10} \phantom{0} \\
 \underline{12} \phantom{0} \\
 12 \overline{) 1} \phantom{0} \\
 \underline{12} \\
 0
 \end{array}$$

$$(237)_{10} = (173)_{12}$$

2

$$50 - (-36) = 50 + 36$$

$$50 = 110010$$

$$36 = 100100$$

$$+50 = 0110010$$

$$+36 = 0100100$$

$$\text{pos} \rightarrow 0110010$$

$$\text{pos} \rightarrow 0100100$$

$$\text{neg.} \rightarrow 1010110$$

Added two positive numbers; answer is negative.

So, Overflow.

3

$$(32112)_5 \times (143)_6$$

$$(32112)_5 \times (223)_5$$

$$32112$$

$$223$$

$$\hline 201341$$

$$1142240$$

$$11422400$$

$$\hline (13322031)_5$$

$$(143)_6 \rightarrow (?)_{10}$$

$$\Rightarrow 1 \times 6^2 + 4 \times 6^1 + 3 \times 6^0$$

$$\Rightarrow (63)_{10}$$

$$(63)_{10} \rightarrow (?)_5$$

$$\begin{array}{r} 5 \overline{) 63} \\ 5 \overline{) 12-3} \\ 5 \overline{) 2-2} \\ 0-2 \end{array}$$

$$(223)_5$$

$$\begin{array}{r} 5 \overline{) 611} \\ 5 \overline{) 11} \\ 1 \end{array}$$